

High Altitude Cerebral Edema (HACE)

Background - High risk with travel to 8,000ft (2,500 m) or higher - May occur at lower altitudes in susceptible population (COPD, CHF, obesity, history of AMS) - Development of symptoms based on: - Rate of Ascent - Sleeping altitude - Strength of hypoxic ventilatory response - Alcohol intake - Obesity - NOT based on physical fitness, age, sex, smoking, previous high-altitude experience - Tend to have recurrence of symptoms whenever they return to the symptomatic altitude - Low partial pressure of O₂ leads to decreased diffusion of O₂ across alveolar spaces --> hypoxemia results in tachypnea and cerebral vasodilation (↓ CO₂) --> increased ICP	Symptoms - Ataxia - Confusion/Altered Mental Status Differential Diagnosis - severe dehydration - hyponatremia - pneumonia - carbon monoxide poisoning - hypoglycemia
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EMT/AEMT

- Follow General Medical Care Guideline
- Perform a full neurologic exam
- Descend immediately** if altered mental status or neurologic deficit
- If practical, Oxygen to improve SpO₂ >90%

EMT/AEMT - Expanded Scope

- Dexamethasone** 8mg PO/IM once if unable to descend immediately

Paramedic

- Follow General Medical Care Guideline
- Descend immediately** if altered mental status or neurologic deficit
- Dexamethasone 8mg ONCE, then 4mg every 6 hours**
- May add **Acetazolamide 250mg BID**, if no contraindication, while descending

****After descent due to HACE, patient should not attempt reascent****

Patient should be evaluated by a physician after descent

Expected SpO₂ and PaO₂ levels at [altitude](#)^[2]

Altitude	SpO ₂	PaO ₂ (mm Hg)
1,500 to 3,500 m (4,900 to 11,500 ft)	about 90%	55-75
3,500 to 5,500 m (11,500 to 18,000 ft)	75-85%	40-60
5,500 to 8,850 m (18,000 to 29,000 ft)	58-75%	28-40

